BHAVYA KUMAR

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EXPERIENCE

ToadEx Inc. Remote

Software Engineering Intern Sept. 2026 - Present

- Designing **Python/FastAPI middleware** for a restaurant kiosk platform with **9 merchant configurations**, separating store-specific payment and printing integrations from Django to simplify maintenance across merchants.
- Traced Vue-to-Django checkout and identified payment-retry and partial-order risks; specified **idempotency keys and SQLite-backed recovery** for interrupted transactions.
- Removed a hardcoded SMTP credential from Django source; validated **4 configuration scenarios** and documented environment-based deployment setup.
- Compared **3 architectural approaches** and proposed REST API contracts and device adapters for TCP/serial terminals and Windows/ESC-POS printers, preserving the existing Vue interface.

EDUCATION

University of Waterloo Waterloo, ON

Honours Bachelor of Mathematics (Co-op) Sept. 2025 - Apr. 2030

Applied Mathematics with Scientific Computing & Scientific Machine Learning

Relevant coursework: Data Structures, Algorithm Design in C, Functional Programming, Probability

Award: President's Scholarship

PROJECTS

Wyag | Git Version Control System May 2026

Python

- Implemented **15 core Git commands** in Python, building Git's object model, a Merkle DAG, and content-addressed storage with SHA-1 hashing and zlib compression.
- Parsed Git's binary index format to support staging, file metadata tracking, and worktree modification detection without Git libraries.
- Debugged serialization and object-identity mismatches and implemented Git-compatible tree ordering to reproduce Git behaviour.

Software Factory | Autonomous LLM Agent Pipeline June 2026

SuperPlane, LLM Agents, Render

- Built a workflow at a New York LLM-agent hackathon to turn natural-language feature requests into software proofs of concept.
- Orchestrated specification, implementation, and verification agents with **validation gates** to prevent failed model output from advancing to subsequent stages.
- Automated code generation, Git commits, and deployment to live Render previews; tested the pipeline against real open issues from the SuperPlane repository.

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, SQL, Racket

Frameworks & databases: Django, Vue.js, React, Node.js, Express, SQLite, Supabase

Tools & technologies: Git, GitHub, REST APIs, LLM APIs, Agent Workflows, Render, Vercel

ACHIEVEMENTS

Bronze Medalist - TIMO International Math Olympiad

Finalist - Northeastern University London Essay Competition